Explanation, Features and Development of LearnHub – Online Learning Platform

1. Introduction

The given code is used to develop a website called LearnHub – Online Learning Platform. It is designed to help students explore different subjects, learn programming, understand web development, study UI/UX design, and develop data skills.

The website is created using HTML5 and CSS3. HTML provides the structure and content, while CSS controls the design, colors, layout, and responsiveness.

LearnHub uses a clean, modern design with a light background, blue highlights, course cards, instructor profiles, learning objectives, and navigation links. It is a front-end website project that demonstrates how an online learning platform interface can be designed.

2. Explanation of the Code

The code consists of two main technologies: HTML and CSS.

A. HTML (HyperText Markup Language)

HTML is used to create the structure of the website. It defines the headings, paragraphs, navigation menu, course cards, buttons, lists, and other content.

HTML tag	Explanation
`<!DOCTYPE html>`	Declares the document as HTML5.
`<html>`	Defines the root element of the webpage.
`<head>`	Contains the page title, metadata, and stylesheet reference.

`<title>`	Sets the browser tab title to LearnHub – Online Learning Platform.
`<body>`	Contains the visible content of the website.
`<header>`	Contains the top navigation area.
`<nav>`	Groups navigation links.
`<main>`	Contains the main content of the webpage.
`<section>`	Organizes the page into separate content sections.
`<article>`	Groups individual course and category cards.
`<div>`	Groups elements for layout and styling.
`<h1>`, `<h2>`, `<h3>`	Display headings and subheadings.
`<p>`	Displays paragraphs and descriptions.
`<a>`	Creates links to different sections of the website.
`<button>`	Displays course action buttons.
`<ul>` and `<li>`	Display lists of learning objectives.

`<footer>`	Displays the website's bottom section.

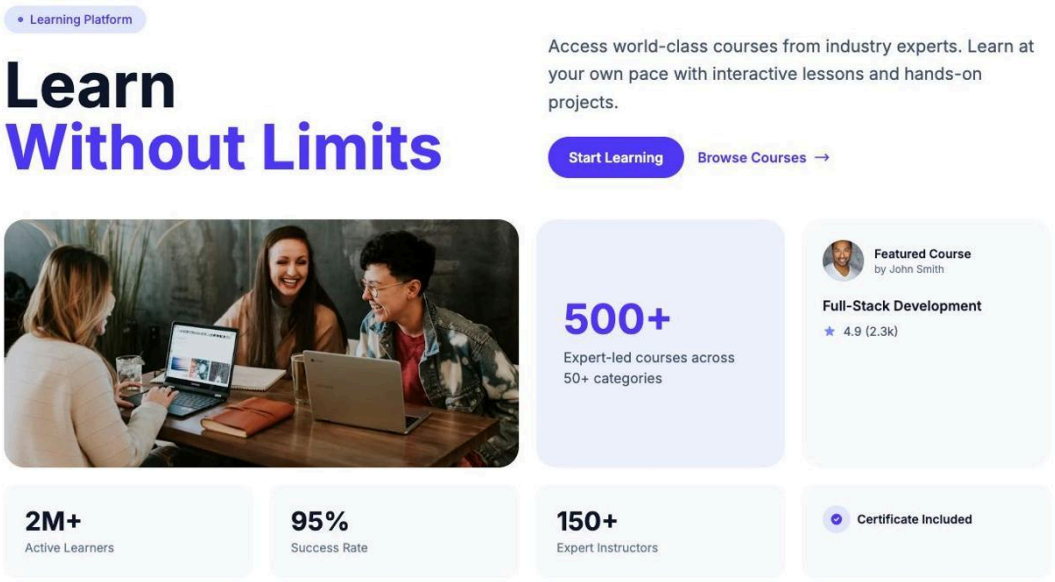
B. CSS (Cascading Style Sheets)

CSS is used to make the website visually attractive. It controls colors, font styles, spacing, borders, card layouts, and responsiveness.

CSS property	Explanation
`background`	Sets the background color of elements.
`color`	Defines text color.
`font-family`	Sets the font used on the webpage.
`margin`	Creates space outside an element.
`padding`	Creates space inside an element.
`display: flex`	Aligns elements horizontally or vertically.
`display: grid`	Arranges content in rows and columns.
`gap`	Adds spacing between elements.
`border-radius`	Rounds the corners of cards and buttons.
`border`	Adds outlines around elements.

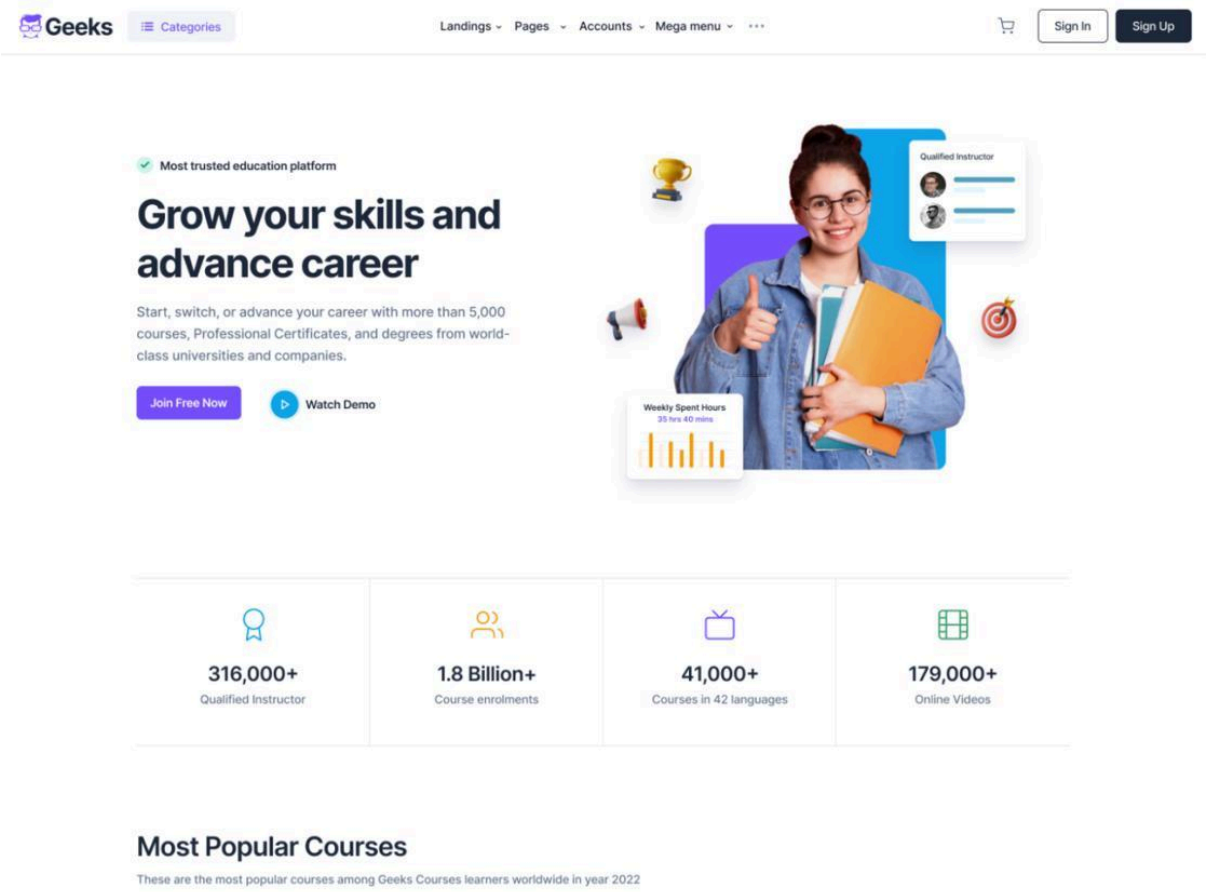
`:hover`	Changes an element's appearance when the pointer moves over it.
`@media`	Changes the layout for different screen sizes.
`scroll-behavior: smooth`	Enables smooth scrolling to linked sections.

3. Features of the Website



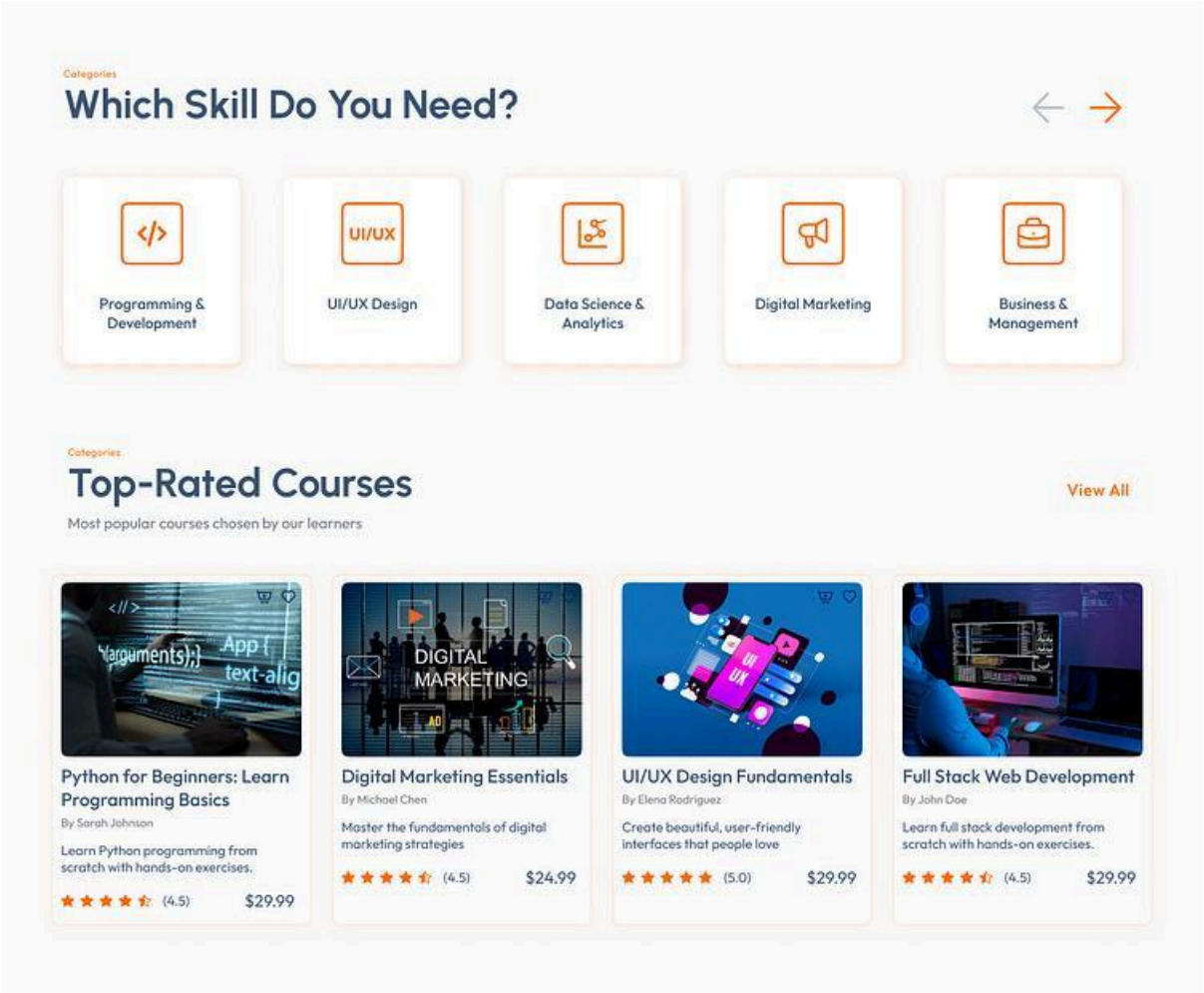
1. Navigation Bar

Displays the LearnHub logo, links to Courses, Categories, Instructors, and About, along with a Browse Courses button. The links help users navigate to relevant sections of the page.



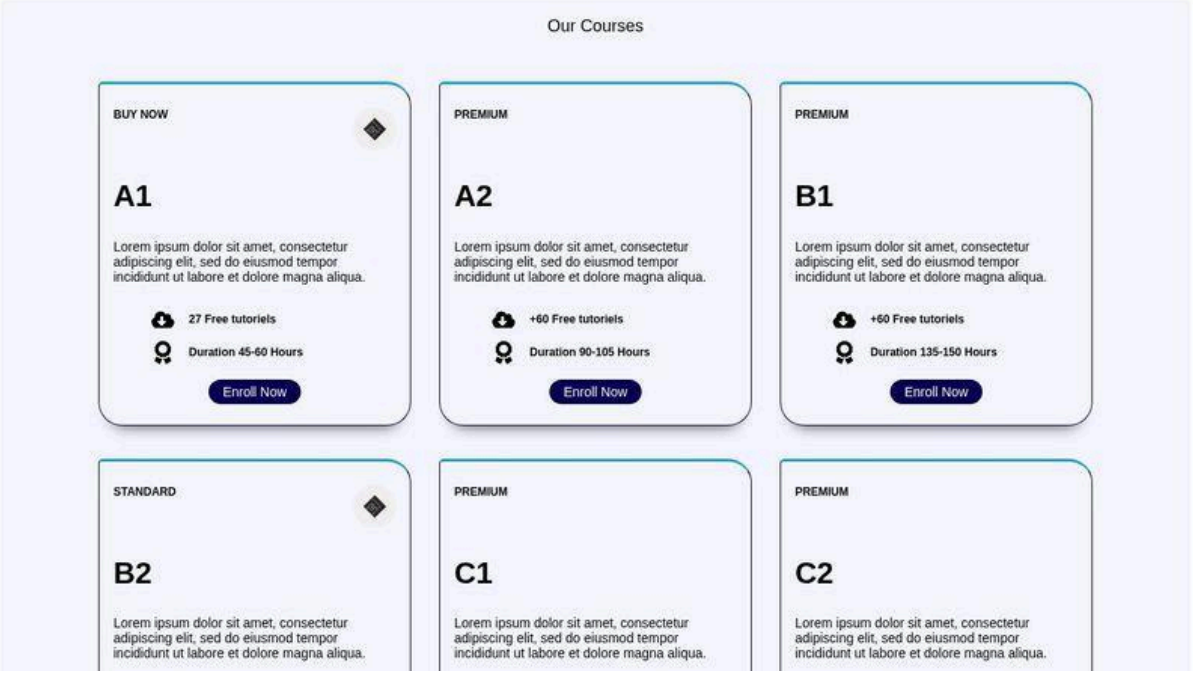
2. Hero Section

Introduces the platform with the message “Build skills for your future.” It includes a short description, Explore Courses and See Learning Goals buttons, and a learning overview showing active courses, completed lessons, learning hours, and certificates.



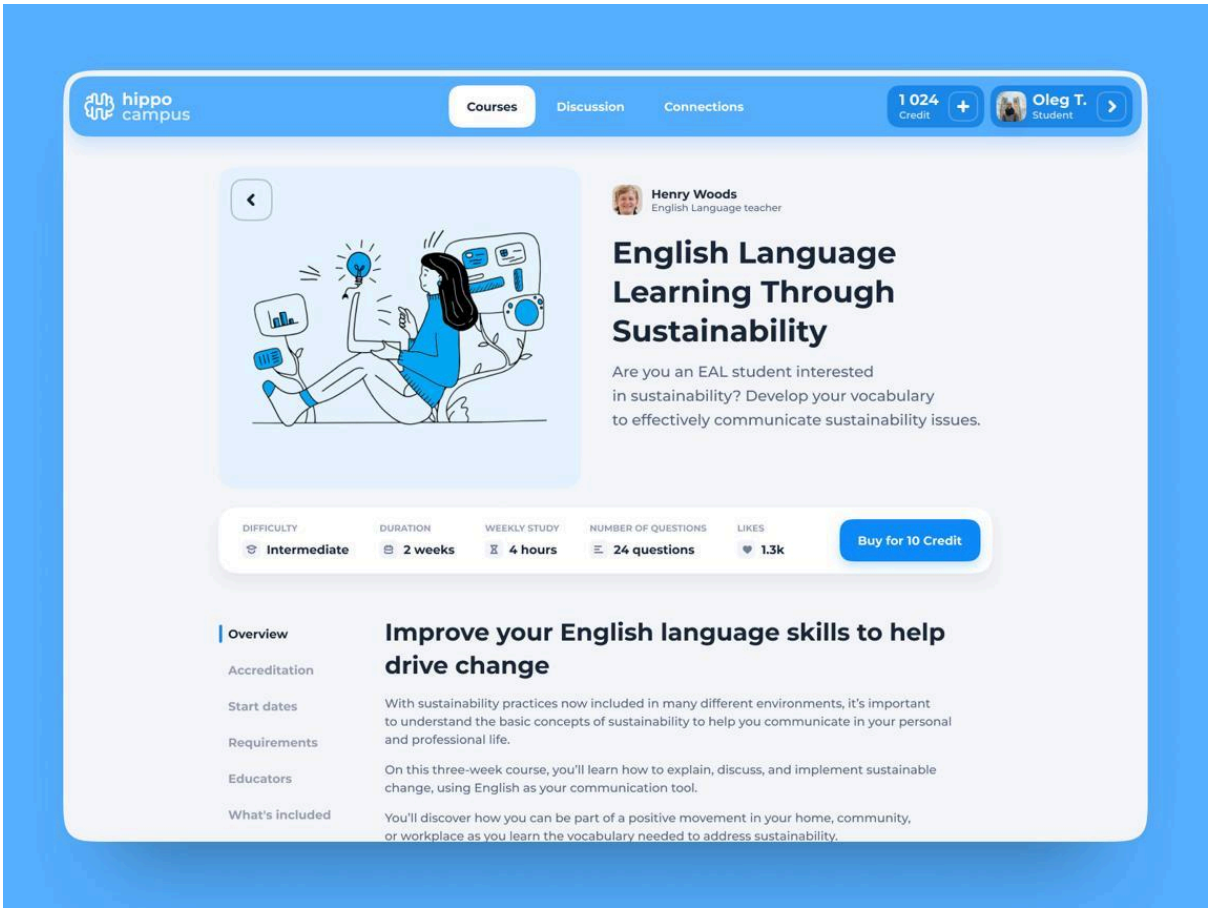
3. Course Categories

Displays four learning categories: Web Development, Programming, UI/UX Design, and Data Skills. Each category contains a title and a short description.



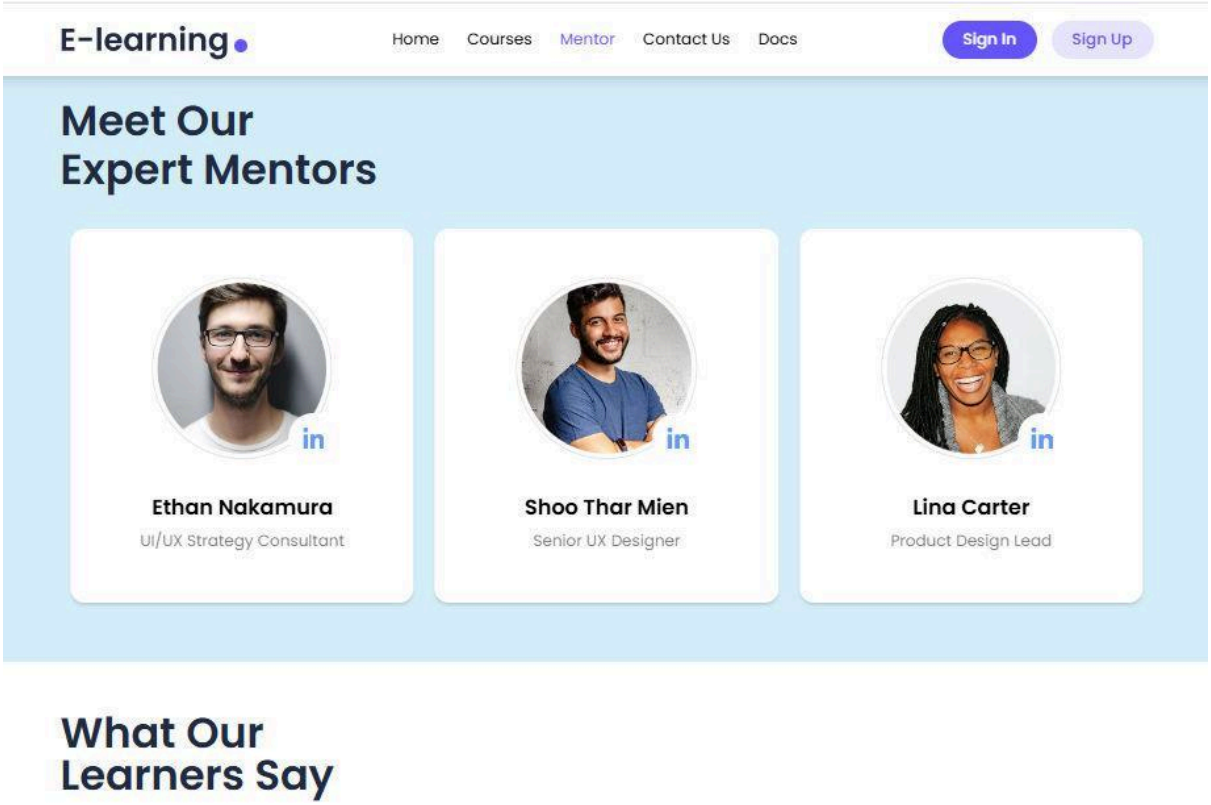
4. Popular Courses

Displays three course cards: HTML & CSS Essentials, Python Programming, and UI Design Fundamentals. Each card includes the difficulty level, number of lessons, course description, instructor details, and an action button.



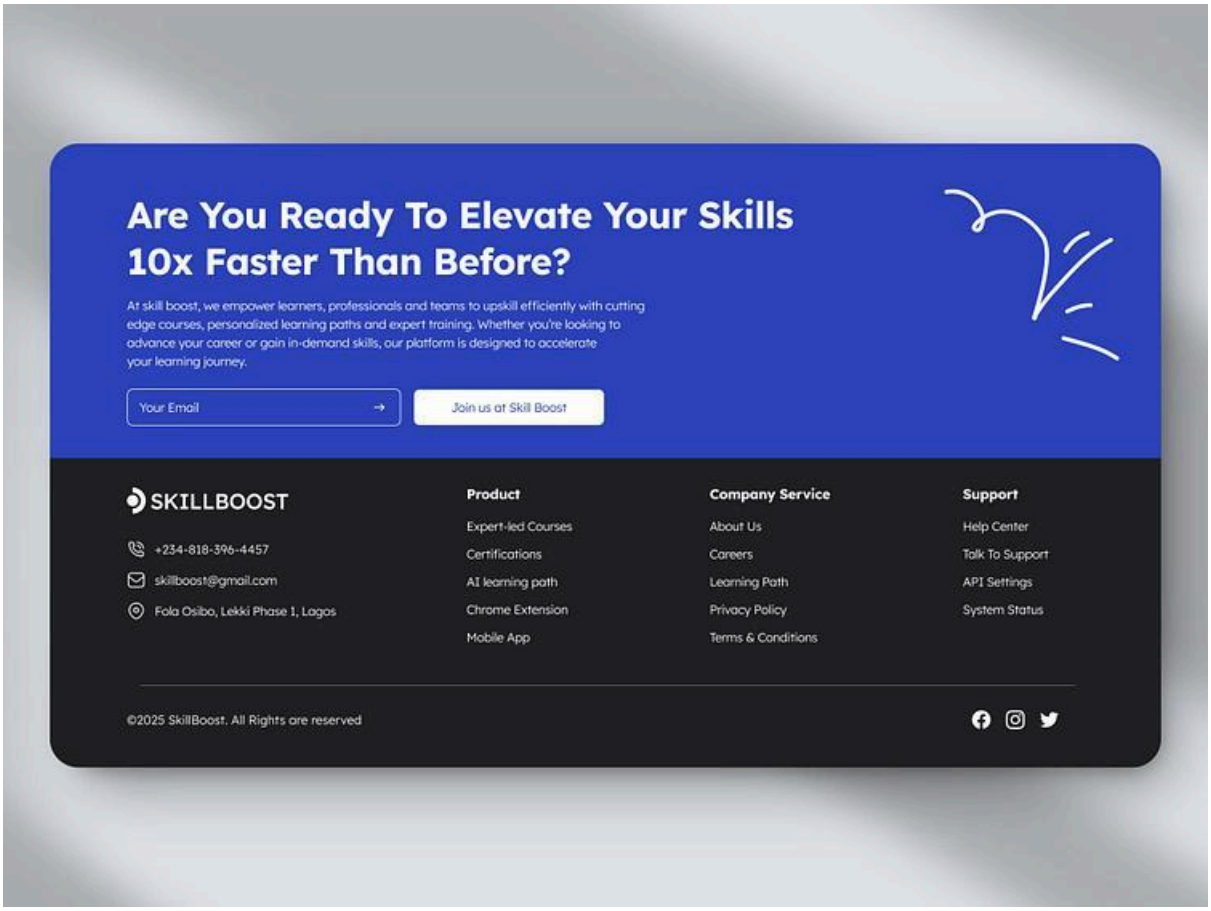
5. Learning Objectives

Explains the learning goals, including understanding concepts, practicing activities, creating small projects, and preparing to study advanced topics.



6. Instructor Profile

Introduces Ananya Kumar as a Web Development Mentor and describes the focus on beginner-friendly HTML, CSS, and responsive web development.



7. About Section and Footer

The About section explains the purpose of LearnHub. The footer displays the platform name, tagline, and copyright information.

Other important features include:

- Responsive design: The layout adjusts to tablets and mobile devices.
- Course information: Shows lesson counts, difficulty levels, and instructor names.
- Interactive styling: Buttons and navigation links have hover or navigation behavior.
- Organized layout: Uses cards, grids, and spacing to make information easy to read.
- Smooth navigation: Internal links scroll to the selected section.
- Reusable CSS classes: Classes such as `.course`, `.category`, and `.info-box` keep the design consistent.

4. Development of the Website

The development process can be explained in the following steps.

Step 1: Create the HTML document

Create a file named `index.html`. Write the basic HTML structure and add the navigation bar, hero section, categories, courses, learning objectives, instructor profile, About section, and footer.

Step 2: Apply CSS styling

Create a file named `style.css` and define the background colors, text styles, spacing, borders, buttons, cards, and navigation appearance.

Step 3: Design the layout

Use Flexbox for navigation, instructor details, and footer alignment. Use CSS Grid to arrange categories, courses, and learning information in columns.

Step 4: Add navigation links

Connect navigation items and buttons to section IDs such as `#courses`, `#categories`, `#instructors`, `#about`, and `#objectives`. This allows users to move between sections.

Step 5: Make the website responsive

Use media queries at 900px and 650px. On smaller screens, change multi-column layouts to fewer columns or a single column, reduce spacing, and hide the navigation links on mobile.

Step 6: Test the website

Open the HTML file in a web browser. Check the section links, card alignment, button appearance, responsiveness, and overall layout. Correct any errors before presenting the project.

5. Folder Structure

The project files can be organized as follows:

LearnHub/

index.html

Contains the website structure and content.

style.css

Contains the design and responsive styles.

Note: Your code includes both a stylesheet link to `style.css` and an internal `<style>` block. For cleaner development, move the CSS into `style.css` and keep the link inside the `<head>` section.

6. Technologies Used

Technology	Purpose
HTML5	Creates the webpage structure.
CSS3	Styles the website and its components.
Flexbox	Aligns items in rows or columns.
CSS Grid	Organizes course and category cards.
Media Queries	Adjusts the design for different screen sizes.
Web Browser	Displays and runs the website.

7. Advantages of the Website

1. Provides a simple interface for exploring learning topics.
2. Organizes courses into clear categories.
3. Displays course levels, lesson counts, and instructor information.
4. Helps students understand learning objectives.
5. Uses a clean and consistent design.
6. Supports different screen sizes through responsive CSS.
7. Demonstrates important HTML and CSS web development concepts.

8. Limitations and Future Improvements

The current LearnHub website is a static front-end interface. The course buttons are styled as interactive controls, but they do not yet enroll students or open actual lessons. The learning statistics are sample display values, not data retrieved from a database.

Future improvements could include:

- Adding JavaScript to handle course button actions.

- Creating student registration and login pages.
- Connecting a database to store student and course details.
- Adding video lessons and course progress tracking.
- Implementing real enrollment and certificate generation.
- Adding a working search and filter option for courses.

9. Conclusion

LearnHub is an online learning platform interface developed using HTML5 and CSS3. It provides organized sections for course categories, popular courses, instructors, learning objectives, and platform information.

This project demonstrates the use of HTML elements, CSS styling, Flexbox, CSS Grid, internal navigation links, and responsive media queries. It is a useful beginner-level web development project that can later be extended into a fully functional e-learning platform.

10. Output

LEARN • PRACTICE • GROW

Build skills for your future.

LearnHub is a simple online learning platform where students can learn programming, design, data skills and more through structured video lessons.

Explore Courses | See Learning Goals

Your learning overview

Active courses	04
Lessons completed	28
Learning hours	16.5h
Certificates	02

Explore categories

Choose a subject and start learning.

</>

Web Development

HTML, CSS, JavaScript and responsive design.

PY

Programming

Python fundamentals and problem solving.

UI

UI / UX Design

Layouts, design principles and user experience.

DS

Data Skills

Data basics, analysis and visualization.

Popular courses

Short lessons with clear learning objectives.

24 Lessons

BEGINNER

HTML & CSS Essentials

Learn the basics of creating clean and responsive websites.

AK **Ananya Kumar**
Web Development Instructor

Continue Learning

30 Lessons

BEGINNER

Python Programming

Understand variables, conditions, loops, functions and lists.

RS **Rahul Sharma**
Python Instructor

Continue Learning

18 Lessons

INTERMEDIATE

UI Design Fundamentals

Create better layouts using spacing, typography and visual hierarchy.

PM **Priya Menon**
UI/UX Instructor

Enroll Now

What you will learn

Every course focuses on practical learning.

Course learning objectives

- Understand important concepts with simple examples.
- Practice each topic through small activities.
- Create small projects using the concepts learned.
- Build confidence to continue with advanced topics.

Meet your instructors

AK

Ananya Kumar

Web Development Mentor

Focuses on beginner-friendly HTML, CSS and responsive web development lessons.

About LearnHub

LearnHub is designed around a simple idea: learning should be clear, organized and easy to follow. Students can choose a course, watch lessons and practice what they learn.

© 2026 LearnHub. Student project interface.

11. Code

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>LearnHub - Online Learning Platform</title>

    <link rel="stylesheet" href="style.css">
</head>
<style>
/* Basic Settings */
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

html{
    scroll-behavior: smooth;
}

body {
    font-family: Arial, Helvetica, sans-serif;
    background: #f7f8fc;
    color: #172033;
    line-height: 1.6;
}
```

```
a {
    text-decoration: none;
    color: inherit;
}

button {
    font-family: inherit;
}


/* Navigation */
.navbar {
    background: white;
    border-bottom: 1px solid #e5e8f0;

    padding: 16px 6%;

    display: flex;
    align-items: center;
    justify-content: space-between;

    gap: 24px;
}

.logo {
    font-size: 24px;
    font-weight: 800;
    color: #2f5bea;
}

.logo span {
    color: #172033;
}
```

```
.nav-links {
    display: flex;
    gap: 28px;

    color: #596273;
    font-size: 15px;
}

.nav-links a:hover {
    color: #2f5bea;
}

.nav-button {
    background: #2f5bea;
    color: white;

    padding: 10px 18px;

    border-radius: 8px;

    font-weight: 700;
}


/* Hero */
.hero {
    padding: 58px 6% 42px;

    display: grid;
    grid-template-columns: 1.3fr 0.7fr;

    gap: 42px;
```

```
    align-items: center;
}

.hero-text .tag {
    display: inline-block;

    background: #eaf0ff;
    color: #2447b8;

    padding: 7px 13px;

    border-radius: 20px;

    font-size: 13px;
    font-weight: 700;

    margin-bottom: 16px;
}

.hero h1 {
    font-size: 48px;

    line-height: 1.1;

    margin-bottom: 18px;

    max-width: 650px;
}

.hero h1 span {
    color: #2f5bea;
}
```

```
.hero p {
    color: #626b7c;

    max-width: 620px;

    margin-bottom: 24px;
}

.hero-buttons {
    display: flex;

    gap: 12px;

    flex-wrap: wrap;
}

.primary-btn,
.outline-btn {
    padding: 12px 19px;

    border-radius: 8px;

    font-weight: 700;
}

.primary-btn {
    background: #2f5bea;
    color: white;

    border: 1px solid #2f5bea;
}
```

```
.outline-btn {
    background: white;
    color: #273247;

    border: 1px solid #d8ddea;
}


/* Hero Card */
.hero-card {
    background: white;

    border: 1px solid #e0e5ef;

    border-radius: 16px;

    padding: 24px;
}

.hero-card h3 {
    margin-bottom: 16px;
}

.stat {
    display: flex;

    justify-content: space-between;

    padding: 13px 0;

    border-bottom: 1px solid #edf0f5;
}
```

```
.stat:last-child {
    border-bottom: none;
}

.stat strong {
    color: #2f5bea;
}


/* Sections */
.section {
    padding: 38px 6%;
}

.section-head {
    display: flex;

    align-items: end;

    justify-content: space-between;

    gap: 20px;

    margin-bottom: 22px;
}

.section h2 {
    font-size: 28px;
}

.section-head p {
    color: #70798a;
```

```
    font-size: 14px;
}


/* Categories */
.categories {
    display: grid;

    grid-template-columns: repeat(4, 1fr);

    gap: 16px;
}

.category {
    background: white;

    border: 1px solid #e0e5ef;

    padding: 20px;

    border-radius: 12px;
}

.category .icon {
    width: 42px;
    height: 42px;

    display: grid;

    place-items: center;

    background: #eaf0ff;
```

```
    color: #2f5bea;

    border-radius: 10px;

    font-weight: 800;

    margin-bottom: 13px;
}

.category h3 {
    font-size: 17px;

    margin-bottom: 4px;
}

.category p {
    color: #727b8c;

    font-size: 13px;
}


/* Courses */
.courses {
    display: grid;

    grid-template-columns: repeat(3, 1fr);

    gap: 22px;
}

.course {
    background: white;
```

```
    border: 1px solid #e0e5ef;

    border-radius: 14px;

    overflow: hidden;
}


/* Course Thumbnail */
.thumbnail {
    height: 170px;

    background: #e8edff;

    position: relative;
}

.lesson {
    position: absolute;

    left: 14px;
    top: 14px;

    background: white;

    padding: 5px 9px;

    border-radius: 6px;

    font-size: 12px;

    font-weight: 700;
```

```
}


/* Course Content */
.course-content {
    padding: 19px;
}

.level {
    color: #2f5bea;

    font-size: 12px;

    font-weight: 800;

    text-transform: uppercase;
}

.course h3 {
    font-size: 19px;

    margin: 7px 0;
}

.course-desc {
    color: #687184;

    font-size: 13px;

    margin-bottom: 15px;
}
```

```
/* Instructor */
.instructor {
    display: flex;

    align-items: center;

    gap: 10px;

    padding-top: 14px;

    border-top: 1px solid #edf0f5;
}

.avatar {
    width: 38px;
    height: 38px;

    border-radius: 50%;

    background: #dfe7ff;

    display: grid;

    place-items: center;

    color: #2f5bea;

    font-weight: 800;
}

.instructor small {
    color: #7a8290;
}
```

```
/* Enroll Button */
.enroll {
    width: 100%;

    margin-top: 16px;

    padding: 11px;

    background: #2f5bea;

    color: white;

    border: none;

    border-radius: 8px;

    font-weight: 700;

    cursor: pointer;
}

.enroll:hover {
    opacity: 0.92;
}


/* Learning Section */
.learning {
    display: grid;

    grid-template-columns: 1fr 1fr;
```

```
    gap: 22px;
}

.info-box {
    background: white;

    border: 1px solid #e0e5ef;

    border-radius: 14px;

    padding: 25px;
}

.info-box h3 {
    margin-bottom: 15px;
}


/* Learning Objectives */
.objectives {
    list-style: disc;

    padding-left: 20px;
}

.objectives li {
    color: #5f6879;

    font-size: 14px;

    margin-bottom: 10px;
}
```

```
/* Instructor Profile */
.instructor-card {
    display: flex;

    gap: 18px;

    align-items: flex-start;
}

.big-avatar {
    width: 70px;
    height: 70px;

    border-radius: 50%;

    background: #dfe7ff;

    color: #2f5bea;

    display: grid;

    place-items: center;

    font-size: 23px;

    font-weight: 800;

    flex-shrink: 0;
}

.instructor-card p {
```

```
    color: #687184;

    font-size: 14px;

    margin-top: 7px;
}


/* About */
.about-text {
    color: #687184;

    font-size: 14px;
}


/* Footer */
.footer {
    margin-top: 30px;

    background: #172033;

    color: #dbe1ed;

    padding: 28px 6%;

    display: flex;

    justify-content: space-between;

    gap: 20px;

    flex-wrap: wrap;
```

```
}

.footer p {
    color: #9fa8b8;

    font-size: 13px;
}


/* Tablet */
@media (max-width: 900px) {

    .hero {
        grid-template-columns: 1fr;
    }

    .categories {
        grid-template-columns: repeat(2, 1fr);
    }

    .courses {
        grid-template-columns: repeat(2, 1fr);
    }

    .learning {
        grid-template-columns: 1fr;
    }
}


/* Mobile */
@media (max-width: 650px) {
```

```
.navbar {
    padding: 14px 5%;
}

.nav-links {
    display: none;
}

.nav-button {
    padding: 9px 13px;

    font-size: 13px;
}

.hero {
    padding: 42px 5% 28px;
}

.hero h1 {
    font-size: 36px;
}

.section {
    padding: 30px 5%;
}

.section-head {
    display: block;
}

.section-head p {
    margin-top: 5px;
}
```

```
    .categories,
    .courses {
        grid-template-columns: 1fr;
    }

    .thumbnail {
        height: 185px;
    }

    .footer {
        padding: 24px 5%;
    }
}
</style>
<body>

    <!-- Navigation -->
    <header class="navbar">

        <a href="#" class="logo">
            Learn<span>Hub</span>
        </a>

        <nav class="nav-links">
            <a href="#courses">Courses</a>
            <a href="#categories">Categories</a>
            <a href="#instructors">Instructors</a>
            <a href="#about">About</a>
        </nav>

        <a href="#courses" class="nav-button">
            Browse Courses
```

```
        </a>

    </header>


    <main>

        <!-- Hero Section -->
        <section class="hero">

            <div class="hero-text">

                <div class="tag">
                    LEARN • PRACTICE • GROW
                </div>

                <h1>
                    Build skills for your
                    <span>future</span>.
                </h1>

                <p>
                    LearnHub is a simple online learning platform
where
                    students can learn programming, design, data
skills
                    and more through structured video lessons.
                </p>

                <div class="hero-buttons">

                    <a href="#courses" class="primary-btn">
                        Explore Courses
```

```
        </a>

        <a href="#objectives" class="outline-btn">
            See Learning Goals
        </a>

    </div>

</div>


<div class="hero-card">

    <h3>Your learning overview</h3>

    <div class="stat">
        <span>Active courses</span>
        <strong>04</strong>
    </div>

    <div class="stat">
        <span>Lessons completed</span>
        <strong>28</strong>
    </div>

    <div class="stat">
        <span>Learning hours</span>
        <strong>16.5h</strong>
    </div>

    <div class="stat">
        <span>Certificates</span>
        <strong>02</strong>
```

```
        </div>

    </div>

</section>


<!-- Categories -->
<section class="section" id="categories">

    <div class="section-head">

        <div>
            <h2>Explore categories</h2>
            <p>Choose a subject and start learning.</p>
        </div>

    </div>


    <div class="categories">

        <article class="category">

            <div class="icon">
                &lt;/&gt;
            </div>

            <h3>Web Development</h3>

            <p>
                HTML, CSS, JavaScript and responsive design.
            </p>
```

```
</article>


<article class="category">

    <div class="icon">
        PY
    </div>

    <h3>Programming</h3>

    <p>
        Python fundamentals and problem solving.
    </p>

</article>


<article class="category">

    <div class="icon">
        UI
    </div>

    <h3>UI / UX Design</h3>

    <p>
        Layouts, design principles and user experience.
    </p>

</article>
```

```
        <article class="category">

            <div class="icon">
                DS
            </div>

            <h3>Data Skills</h3>

            <p>
                Data basics, analysis and visualization.
            </p>

        </article>

    </div>

</section>


<!-- Popular Courses -->
<section class="section" id="courses">

    <div class="section-head">

        <div>
            <h2>Popular courses</h2>

            <p>
                Short lessons with clear learning objectives.
            </p>
        </div>
```

```
</div>


<div class="courses">

    <!-- Course 1 -->
    <article class="course">

        <div class="thumbnail">

            <span class="lesson">
                24 Lessons
            </span>

        </div>


        <div class="course-content">

            <span class="level">
                Beginner
            </span>

            <h3>
                HTML & CSS Essentials
            </h3>

            <p class="course-desc">
                Learn the basics of creating clean
                and responsive websites.
            </p>
```

```
        <div class="instructor">

            <div class="avatar">
                AK
            </div>

            <div>
                <strong>Ananya Kumar</strong>

                <br>

                <small>
                    Web Development Instructor
                </small>
            </div>

        </div>


        <button class="enroll">
            Continue Learning
        </button>

    </div>

</article>


<!-- Course 2 -->
<article class="course">

    <div class="thumbnail">
```

```
    <span class="lesson">
        30 Lessons
    </span>

</div>


<div class="course-content">

    <span class="level">
        Beginner
    </span>

    <h3>
        Python Programming
    </h3>

    <p class="course-desc">
        Understand variables, conditions,
        loops, functions and lists.
    </p>


    <div class="instructor">

        <div class="avatar">
            RS
        </div>

        <div>
            <strong>Rahul Sharma</strong>

            <br>
```

```
                <small>
                    Python Instructor
                </small>
            </div>

        </div>


        <button class="enroll">
            Continue Learning
        </button>

    </div>

</article>


<!-- Course 3 -->
<article class="course">

    <div class="thumbnail">

        <span class="lesson">
            18 Lessons
        </span>

    </div>


    <div class="course-content">

        <span class="level">
```

```
    Intermediate
</span>

<h3>
    UI Design Fundamentals
</h3>

<p class="course-desc">
    Create better layouts using spacing,
    typography and visual hierarchy.
</p>


<div class="instructor">

    <div class="avatar">
        PM
    </div>

    <div>
        <strong>Priya Menon</strong>

        <br>

        <small>
            UI/UX Instructor
        </small>
    </div>

</div>


<button class="enroll">
```

```
                Enroll Now
            </button>

        </div>

    </article>

</div>

</section>


<!-- Learning Objectives -->
<section class="section" id="objectives">

    <div class="section-head">

        <div>

            <h2>
                What you will learn
            </h2>

            <p>
                Every course focuses on practical learning.
            </p>

        </div>

    </div>


    <div class="learning">
```

```
<div class="info-box">

    <h3>
        Course learning objectives
    </h3>

    <ul class="objectives">

        <li>
            Understand important concepts
            with simple examples.
        </li>

        <li>
            Practice each topic through
            small activities.
        </li>

        <li>
            Create small projects using
            the concepts learned.
        </li>

        <li>
            Build confidence to continue
            with advanced topics.
        </li>

    </ul>

</div>
```

```
<!-- Instructor Profile -->
<div class="info-box" id="instructors">

    <h3>
        Meet your instructors
    </h3>

    <div class="instructor-card">

        <div class="big-avatar">
            AK
        </div>

        <div>

            <h3>
                Ananya Kumar
            </h3>

            <strong>
                Web Development Mentor
            </strong>

            <p>
                Focuses on beginner-friendly
                HTML, CSS and responsive
                web development lessons.
            </p>

        </div>

    </div>
```

```
            </div>

        </div>

    </section>


    <!-- About -->
    <section class="section" id="about">

        <div class="info-box">

            <h3>
                About LearnHub
            </h3>

            <p class="about-text">
                LearnHub is designed around a simple idea:
                learning should be clear, organized and easy
                to follow. Students can choose a course,
                watch lessons and practice what they learn.
            </p>

        </div>

    </section>

</main>


<!-- Footer -->
<footer class="footer">
```

```
        <div>

            <strong>
                LearnHub
            </strong>

            <p>
                Simple learning. Useful skills.
            </p>

        </div>



        <p>
            © 2026 LearnHub. Student project interface.
        </p>

    </footer>



</body>
</html>
```